

Comparative Advances in Financial Sentiment Analysis: A Review of BERT, FinBert, and Large Language Models

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Abstract— Capturing market sentiments and supporting well-informed financial decision-making depend on the developing field of Financial Sentiment Analysis (FSA). Natural language processing (NLP) has made significant strides in comprehending and categorizing sentiment in intricate financial texts, especially with the use of Large Language Models (LLMs). The application of various LLMs, such as Bidirectional Encoder Representations form Transformers (BERT) and Financial BERT (FinBERT), as well as distilled models, such as DistilBERT and DistilRoBERTa, on a variety of financial datasets, including Financial Phrase Bank and LexisNexis news articles, was the main focus of this review article, highlighting different approaches such as model fine-tuning, zero-shot and few-shot learning, and prompt engineering. The study focuses on practical model predictions through a case study of sentimental analysis of cryptocurrencies. While FinBERT, a financial variant of BERT, exhibits high accuracy and robustness, other LLMs exhibit varying degrees of success based on the dataset and domain requirements. The analysis concentrates on the difficulties, compromises, and potential paths for improving LLMs for financial sentiment analysis.

Keywords—Fin Bert; LLMs; BERT; Financial Sentimental Analysis; FSA; Financial Datasets; Data preprocessing;

I. INTRODUCTION

Sentiment analysis is a critical application of Natural Language Processing (NLP) that classifies text as neutral, negative, or positive. In finance, it helps understand the attitudes of consumers and investors by analyzing data from diverse sources such as news articles, financial reports, and social media. This approach enables predictions about market movements[1],[2]. For instance, positive sentiment about a company on social media may indicate a potential rise in its stock price, whereas negative sentiment could

signal a downturn. Active learning allows sentiment classifiers to adapt to real-time financial data, improving their accuracy and relevance in a dynamic market environment[3].

Granger causality, a statistical hypothesis test, determines whether one time series can predict another. It is often used to explore the relationship between sentiment and stock price changes[4]. Sentiment data from social media, which is inherently noisy and unstructured, is processed using machine learning models like Support Vector Machines (SVMs). These models are effective in converting large volumes of unstructured data into actionable insights. This capability is especially valuable in the financial industry, where sentiment-driven market changes can occur rapidly[5][6].

Sentiment analysis extends beyond individual stock predictions, offering broader applications such as economic forecasting. Public sentiment indicators derived from social media can reveal optimism or pessimism about the economy, aiding decision-making for both investors and policymakers. Financial institutions use sentiment analysis to manage risks, refine strategies, and anticipate market trends effectively.

Large Language Models (LLMs) like GPT (Generative Pre-trained Transformer), Bidirectional Encoder Representations from Transformers (BERT), and their specialized variants, including BloombergGPT and Financial Bidirectional Encoder Representations from Transformers (FinBERT), significantly enhance sentiment analysis by processing large-scale financial data with precision. These models analyze data from social media, financial reports, and news articles, providing real-time insights into market trends, stock performance, and investor

sentiment. Unlike traditional models, LLMs use sophisticated neural network architectures to identify subtle patterns, making them highly effective for analyzing complex financial datasets. Domain-specific models like FinBERT address the nuanced meanings of financial terminology, improving sentiment analysis accuracy[7].

Adaptation techniques such as few-shot learning and transfer learning further enhance LLMs' ability to process evolving financial jargon and market trends. However, raw financial data often contains issues like null values, irrelevant content, and improper structures. Section 4 of this review discusses data preprocessing techniques that transform such data into a usable format, ensuring reliability in subsequent analyses.

Section 5 explores the use of LLMs for cryptocurrency market prediction, where their adaptability to blockchain data and crypto-specific jargon is crucial. Cryptocurrency markets are volatile and complex, requiring models capable of analyzing technical language and transaction logs[8]. Crypto-specific LLMs can detect fraud, predict price movements, and offer enhanced security, outperforming standard models that struggle with specialized language.

challenges unique to financial NLP, especially model interpretability, normative considerations, and computational demands in the models themselves, to provide glimpses into future research and development directions for this evolving field. This article carries out a comparative analysis to guide researchers and practitioners in the selection and optimization of LLMs to improve sentiment-based decision-making in financial markets.

II. EVOLUTION OF FINANCIAL SENTIMENTAL ANALYSIS TECHNIQUES

Traditional approaches to sentiment analysis utilized dictionary-based methods, for example, Loughran-McDonald's lexicons, to classify financial terms as positive or negative. These interpretable methods were accurate in rapidly changing markets but failed to capture the complexities of financial language. Machine learning models such as Naive Bayes and SVMs gained popularity because they were effective with large datasets, but they offered limited feature extensions and lagged behind in understanding nuanced financial contexts[9]. Rule-based techniques, coupled with NLP tools like Part-of-Speech tagging, improved robustness by aiding context disambiguation. The Time line of the LLMs is shown in Fig 1.

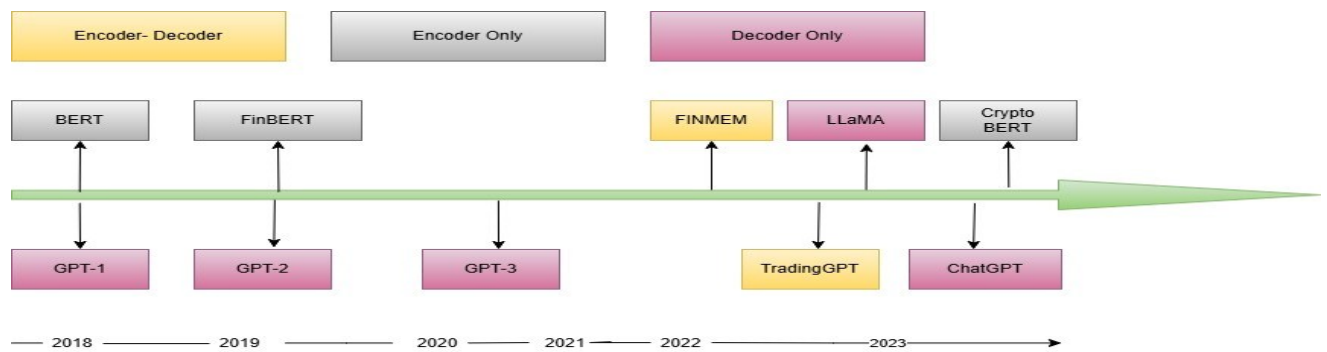


Fig. 1. Timeline of LLMs and their classifications.

Looking ahead, the integration of multimodal LLMs could revolutionize financial sentiment analysis. These models would combine textual data with images, videos, and transactional information to provide a comprehensive view of market sentiment. For instance, they could analyze news articles, social media posts, and visual charts simultaneously to predict market behavior.

Ethical challenges remain significant. Issues such as preventing market manipulation and ensuring compliance with privacy laws necessitate ongoing advancements in model development. Balancing innovation with ethical and legal considerations will be crucial to unlocking the full potential of LLMs in the financial sector.

In an attempt to present an expansive analysis of the latest developments in the area of financial sentiment analysis, this paper aims to explore them specially through the application of large language models like BERT, FinBERT, and their derivatives. This article tries to bring to the fore the potential benefits, drawbacks, and particular applications of LLMs while providing an overview of how these models interpret and classify financial sentiment across various data sources like news articles, social media, and earnings reports. These models examine some of the

Deep learning brought a major advancement, and models such as RNNs(Recurrent Neural Network) and LSTMs(Long Short-Term Memory) started using text dependencies to parse sentiment in complex financial documents. Their sequential memory was useful for tasks where word order impacts the detection of sentiment, like analyzing financial news[10]. However, their weakness with large datasets hindered real-time applications. Transformers like BERT and GPT changed the face of sentiment analysis by scanning it bidirectionally, thus capturing the contextual meaning much better than ever. FinBERT is a finance-specific variant of BERT fine-tuned on financial texts that enhance market sentiment forecasting and anomaly detection. Such models outperform traditional approaches, especially when dealing with noisy and linguistically varied data from sources like social media.

Combining event detection with sentiment analysis, such real-time applications are capable of handling high-frequency data like that of Twitter and associating changes in sentiment with events of the market. It is ensured that these models constantly adapt to market volatility in domains like cryptocurrency through active learning. Despite these advances, high accuracy across markets and languages remains a challenge. Financial sentiment models

TABLE I. Overview of Financial Sentimental analysis Models

Model	Training Data	Key Characteristics	Advantages	Limitations
BERT	General language corpora (Wikipedia, Book Corpus)	Transformer-based; bidirectional text understanding	High accuracy for general NLP tasks; contextual understanding	Needs fine-tuning for domain-specific language
FinBERT	Financial Phrase Bank, LexisNexis news articles	Fine-tuned on financial data; finance-specific vocabulary	High accuracy in financial sentiment; domain-specific	Limited to financial texts; high computational cost
DistilBERT	Distilled from BERT, with reduced parameters	Lightweight, faster version of BERT	Lower resource requirements; good for low-latency tasks	Minor accuracy drop compared to BERT
GPT	Diverse internet-based data and fine-tuning corpora	Few-shot learning; prompt-based; large model size	Handles complex financial narratives; adaptable	Computationally intensive; requires fine-tuning
DistilRoBERTa	Distilled version of RoBERTa, trained on general corpora	Optimized for speed; handles informal language	Suitable for noisy data (e.g., social media); faster inference	Less robust than RoBERTa on some NLP tasks

require regional or linguistic adaptation; financial terminology differs in meaning by context. Finance professionals also need interpretability, so SHAP(SHapley Additive exPlanations) and attention mechanisms will be important for explaining the predictions of sentiment[11].

BERT revolutionized the NLP with new types of bidirectional training, particularly improving tasks such as sentiment classification and text inference. It pushed it to the next level and, with pre-training on texts specifically in finance, provided a nuance detection sensitivity toward financial sentiment. Applications range from predicting stock movements through risk assessment and market sentiment interpretation. Other integrations include the processing of numerical and categorical information to enhance credit prediction using text feature extraction with FT-Transformer rather than the use of just traditional models like Random Forest and XGBoost[12-16]. Modern innovates are distilled models zero-shot and few shot and prompt engineering addressing various aspects such as fewer or rather no labeled data together with high computational cost distileds cut the same model which seems to retain a state or performance, in these context, ideal for any scenario of low resources whereas domain unrelated training will take off that phenomenon of zero shot with better adaptation in specific Lexions.

The zero shot method enhances the model's performance on time-specific sentiment, which is an important requirement in dynamic fields like cryptocurrency. All these innovations help form financial sentiment analysis by rendering the models not only efficient but also interpretable. All these techniques finally converge toward scalable adaptive solutions that are sure to present timely insight from huge diversified data repository in ever-evolving markets to the financial houses and traders.

The Overview of different Financial Sentimental Analysis(FSA) Models is given below in Table I.

III. LARGE LANGUAGE MODELS FOR FINANCIAL SENTIMENTAL ANALYSIS

The Language model's pipeline for sentimental analysis is explained in the Fig1. One of the significant LLM for Sentimental Analysis is BERT. The BERT architecture has overturned the natural language processing in many diverse areas, including finance. The fact that BERT can perform bidirectional training means that both words preceding and those following a word in a sentence could be taken into consideration, which is a very important feature in understanding complex language structures. The BERT architecture is based on Transformer models with multiple layers of self-attention mechanisms capturing long-range dependencies across tokens. This architecture empowers BERT to perform outstandingly at tasks involving the fine-grained understanding of language, like sentiment analysis and question-answering tasks. However impressive that is to pre-train BERT on general corpora such as the contents of Wikipedia and BookCorpus, its capabilities cannot be transferred directly to the technical language of finance[17].

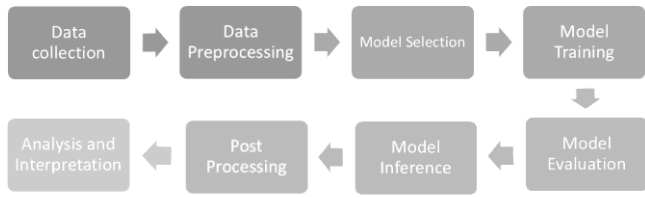


Fig 2: LLM Pipeline for Sentiment Analysis

FinBERT is a variant of BERT replacing the pre-training tasks by MLM and NSP, with tasks such as Span Replace Prediction and Capitalization Prediction. FinBERT goes outstandingly beyond the state-of-the-art models over datasets such as Financial PhraseBank and FiQA(Financial Question Answering) with very high efficiency in carrying out aspect-based sentiment analysis-a task critical to grasping how one feels towards relevant entities in a given piece of financial literature[18]. FinBERT improves the quality of financial question answering and sentence boundary detection. That is critical for complex regulatory and legal documents processing, but that has high computation requirements, so it does not have availability in many resource-constrained settings and hence, the adopted models are DistilBERT and DistilRoBERTa.

Hugging Face developed DistilBERT, retaining 97% of the capabilities of the BERT language while coming with half the parameters, hence very property-suitable for low-latency applications like algorithmic trading and market monitoring[20]. Although less accurate, its speed and efficiency with resources make it suitable for use in applications that are oriented towards speedy processing. The DistilRoBERTa further improves the management of diverse linguistic patterns with consistent financial sentiment analysis results. Such distilled models can be used on standard hardware, thereby democratizing the use of advanced NLP tools by small businesses and individual traders.

The distilled models are always a compromise of efficiency and performance. Although the lower accuracy may not have very good implications for most the application areas of deep linguistic understanding, the distilled versions do seem to work properly for high-throughput application of real-time sentiment analyses. The higher complexity types of applications are better catered to by larger sizes. For example, Large Language Models, in LLaMA and ChatGPT, fine-tune with a view of functionalities: the ability to monitor sentiment specifically in real-time or its application in sentiment-driven trading strategies[21]. There are actionable insights of nuances that trends display in terms of the sentiment from unstructured data toward fine-tuning trading strategies. LLMs are more potent as compared to simple models.

LLMs are trained on many datasets so they pick the minute sentiment cues that exist in social media and news channels. There is also real-time deployment for monitoring purposes to shape strategies according to trader sentiments. They also score the intensity of volatility along with the short-term forecast of price movements. Both FINMEM and TradingGPT work with memory modules and multiple agent systems that can make strategy dynamic changes based upon historical as well as time-series sentiment trends[8]. A system would simulate a portfolio of strategies that would broaden the choices in decision making about different insights from sentiment.

Potential exceptions, such as relatively high computing needs and generally lower transparency. Their "black box" nature together with the intensity of use of the resources makes availability to such tools difficult by smaller firms. Synthetic datasets add further to sentiment-based trading with simulated diversified market conditions[19]. With the interpretation of better models and optimization techniques coming their ways for more general usage, reliability of models increases. The problems persist, but LLMs are revolutionizing the world of financial sentiment analysis in the direction of trading strategies.

IV. DATASETS AND PREPROCESSING FOR FINANCIAL SENTIMENTAL ANALYSIS

A. Financial Datasets

Financial phrase sentiment analysis employs datasets such as Financial Phrase Bank, LexisNexis, and TRC2-financial. These datasets differ in the analytical needs that they provide. Financial Phrase Bank was developed by Malo et al. (2013) that comprises 4,845 annotated sentences, which were mainly gathered from news from the LexisNexis[22]. Each sentence was assigned a label either positive, negative, or neutral according to the effect of the sentence on the stock price. This consensus-based annotation is reliable, and the dataset is a great one for supervised learning models that have to classify very faint sentiments in the financial language. Table II illustrates how its structured annotations make sense of the terms like "debt" and "loss" for better understanding of financial jargon.

The following table III illustrates the comparison of some popular financial datasets used for training financial models. The table contains an example of their structure.

TABLE II. Samples of Some Financial Datasets

Dataset	Example
Phrase Bank	“sentence”: “Pharmaceuticals group Orion Corp reported a fall in its third-quarter earnings that were hit by larger expenditures on R&D and marketing .”, “label”: “negative”
SemEval	“id”: 2, “company (target)”: “Morrisons”, “title”: “Morrisons book second consecutive quarter of sales growth”, “sentiment”: 0.43
FiQA	“sentence”: “Royal Mail chairman Donald Brydon set to step down”, “info”: [“snippets”: [“set to step down”]], “target”: “Royal Mail”, “sentiment score”: “-0.374”, “aspects”: [“Corporate/Appointment”]]

The LexisNexis database is a vast collection of unstructured financial data: news, legal filings, and corporate earnings[23]. The database is deep in timeframes, regions, and industries, which make it a good source for the creation of custom sentiment datasets. It has to be preprocessed, categorized, remove duplicates, and label according to domain-specific sentiments. It also captures shifts in sentiment based on the changes in the global or regional economies.

The TRC2-financial dataset comes from Reuters TRC2 news and comprises 46,143 financial documents between the years 2008 and 2010; there are 29 million words and 400,000 sentences[24]. The domain-orientated dataset allows a model like FinBERT to learn some nuances of language in finance - its ability to understand how sentiment shifts around corporate earnings or macroeconomic trends. Thereby, the comprehensive scope will enable deep learning models to predict the risks and invest precisely. These datasets, through continuous updates and refinements, are important for furthering real-time sentiment analysis that keeps financial models relevant in dynamic markets.

TABLE III. COMPARISON OF FINANCIAL PHRASEBANK, LEXISNEXIS AND TRC2 – REUTERS NEWS

Datasets	Size	Sentimental Categories	Annotation Quality
Financial Phrase Bank	~4,845 sentences	Positive, Negative, Neutral	High
LexisNexis	Millions of Documents	Customizable	Moderate

B. Data Preprocessing

Data Preprocessing is a very important step, wherein raw financial text is converted into structured data for models like FinBERT. The preprocessing steps are explain in the Fig 3. Financial datasets are cluttered with unstructured data mostly, which needs cleaning so that noise from HTML tags, advertisements, and redundant content is removed which is called data cleaning[25]. This makes sure that the models focus on meaningful financial terms and sentiments and maintain domain-specific language as small terminological changes can drastically change the interpretation of sentiment.

Tokenization further breaks the text into even smaller units. In financial NLP, tokenization preserves multi-word terms like "interest rate hike" because they are treated as one word and therefore preserve their contextual meaning. Industry-specific terms, such as ticker symbols and abbreviations, for example "IPO"(Initial Public Offering) or "EPS"(Earnings Per Share), are preserved for analytical purposes.

Data balancing removes biases in sentiment datasets, which favor general classes. It could be either the positive or neutral class. Over-sampling of less represented classes will ensure that models generalize well and catch sensitive sentiment indicators such as rare yet critical risks.

Named Entity Recognition (NER) and entity linking identify the most important financial entities, including companies and instruments. The sentiment labels that tag the entities enable pinpointing which entities are associated with market sentiment shifts[25]. These domain-specific lexicons improve NER to handle ambiguities in the company names-for example, "Apple" or "Tesla"-to attain contextual accuracy .

Handling sarcasm and figurative language is another challenge, particularly in social media texts. Advanced techniques for sentiment disambiguation analyze tonal cues, context, and user intent, so models can detect sentiments masked by rhetorical expressions.

In essence, data normalization standardizes the representations for currency, date, and so on. Other applied techniques include deduplication, distributed processing, and even data compression. All these preprocessing steps ensure that financial models deliver timely, accurate, and reliable insights necessary for risk assessment and decision making.

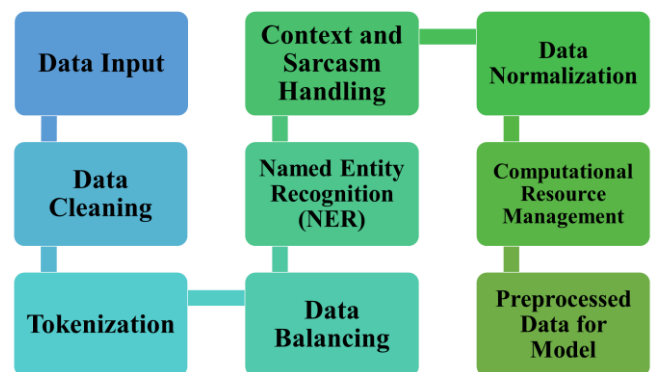


Fig 3:Data Preprocessing steps

V. CASE STUDY: CRYPTOCURRENCY MARKET PREDICTION USING LLMs

Analysis of news sentiment is an integral part of studying trends for the cryptocurrency market as public opinion significantly influences market dynamics. Volatility in the cryptocurrency sector then is created through the interplay of both technical or regulatory events and shifts in investor psychology, which in turn are modified by real-time judgments of sentiment. The integration of news sentiment with market indicators such as price, volume, and volatility allows one to extract insights about the effects of positive, neutral, or negative news on investor behavior and market trends[26 -28]. For instance, if there is positive news on Bitcoin or Ethereum, that is, institutional investments or regulatory body endorsements, the trading increases, and prices tend to rise because of increased confidence among investors.

Advanced NLP models such as BERT, FinBERT, or even ChatGPT are useful in making subtle analyses over news sentiment as it understands what complex financial language is conveyed to thusly allow for further insight on how public discourse influences markets[29]. Given all of this is taking place online, their models allow for real-time insight over the sentiment of investors who are affected by sentiment and not by fundamentals at least over the short run. Behavioral finance theories focus on how news creates "herd behavior" where bad news such as the regulatory crackdown will spur selling and good news, like improved technology, will spur a rally in price.

Time-series analysis further enriches cryptocurrency sentiment analysis by correlating historical sentiment trends with market data. For example, increased positivity toward Bitcoin often forecasts price surges within hours. Regional sentiment analysis also shows how local regulations and economic factors influence investment behaviors, allowing for more precise predictions and market strategies.

LLMs like GPT-4, fine-tuned specifically for financial sentiment, support the use of real-time monitoring tools and predictive algorithms. These models are optimized by advanced optimizers like ADAMW, fine-tuning hyperparameters such as learning rates, and backtested to ensure robust predictions[30],[31]. Tools using FinBERT or similar models offer actionable insights, which supports risk assessment and enables proactive decision-making within the volatile cryptocurrency market. In this regard, these advances further cement the position of sentiment analysis in guiding investor strategy and understanding market trends.

VI. CHALLENGES AND FUTURE DIRECTIONS

Advanced NLP models in financial sentiment analysis introduce disturbingly elementary ethical concerns, such as the bias in the training data. The bias will be perpetuated with the financial sentiment models derived from enormous amounts of textual data culled from news, social media, or reports. Suppose a model was trained mainly on Western financial news and misinterprets the sentiments of a non-Western context, making bad analyses and poor decisions.

Incorporating large language models like BERT into financial sentiment analysis presents a host of challenges including model bias, lack of interpretability, high computational requirements, and ethical concerns around data privacy and market manipulation[32-34]. Trust requires transparency, but models such as BERT are a "black box" in which decision-making processes cannot be easily understood. Such lack of clarity can lead to misinterpretations, which could influence the most critical decisions made in markets and thereby increase volatility.

LLMs are computationally expensive to train and infer and therefore very energy-intensive and environmentally unfriendly. Such a cost is hard for small firms and startups to bear when the importance of sustainable technology increases. The trade-offs are very important between model complexity and efficiency, especially for firms relying on high-frequency trading, requiring very low latency and high precision predictions.

Domain-specific adaptation is important because financial language is very rich in jargon and subtle expressions. While pre-trained models like FinBERT do exceptionally well when fine-tuned for financial data, they require constant updates to reflect changes in language, regulations, and market conditions. This requires a lot of resources, which becomes challenging for smaller organizations.

All of these models will require ethical considerations, computational cost, and adaptability as a guideline in their use for responsible and equitable benefits towards financial markets. Transparency, reduction of bias, and compliance with regulation, such as GDPR(General Data Protection Regulation), is critical for building trust and integrity within financial sentiment analysis.

VII. CONCLUSION

Innovations within financial markets, propelled by the incorporation of machine learning and correlation-centric indexing, are redefining investment approaches and enhancing risk management practices. Conventional capitalization-weighted indices frequently skew performance evaluations by disproportionately highlighting large-cap stocks; in contrast, correlation-based indices provide a more equitable depiction of sector interactions, thereby enhancing benchmarking accuracy and fostering stability. Advanced machine learning frameworks like CNN-LSTM improve analytical capacity significantly because these frameworks can effectively capture non-linear patterns in stock fluctuation and market signals, which helps investors better identify trends and optimize their returns while at the same time reducing risk. Moreover, contemporary methodologies, such as HRP(Hierarchical Risk Parity) portfolio optimization, are aligned with these indices, providing powerful solutions for risk management, especially in turbulent markets, thus motivating sustainable investing practices.

Advanced indexing at the corporate level enables organisations to better measure performance thus allowing for well-informed strategic decision-making. On the other hand, machine learning frameworks provide greater market

visibility as well as compliance monitoring, even though "black box" traits of these frameworks illustrate the need for explaining AI to sustain trust and accountability. Emerging trends in multimodal analysis, integrating data about executive tone and financial media visuals, promise a better understanding of market sentiment, advancing both finance sentiment analysis and model evaluation. Altogether, innovations improve the state of the representation of the market, its risk management, and its forecasting capabilities, fortifying a more resilient financial system capable of adaptation in a complex and evolving economic landscape.

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